Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

Bachelor in Computer Applications

Course Title: **Digital Logics**

Code No: Semester: **1**

Full Marks: **60**   Pass Marks: **24**   Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

***Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

1. Design a BCD-to Seven-segment decoder to display decimal numbers 1, 2 and 4.

2. What are the drawbacks of clocked RS flip flop? Explain the operation of a JK flip flop along with its characteristic table, characteristic equation circuit diagram and timing diagram.

3. Design a sequential circuit using JK flip-flop with the help of given state diagram.

4. Convert (110.101)8into binary and decimal number system.

5. Subtract (739.57)■■— (78.35)■ using both 10's and 9's complement.

6. What is magnitude comparator? Design 2-bit magnitude comparator.

7. Define Half-subtractor with truth table and logic diagram.

8. What is decoder? Implement 8 × 1 MUX using 2 × 1 MUX.

9. Design and explain the operational characteristics of Dflip flop.

10. Design Mod-5 synchronous counter using T- flip flop.

11. Draw a Serial-In Serial-Out Shift register and explain it.

12. Write short notes on: (Any two)a. BCD codeb. Status Registerc. Ring Counter